

Development of Solid Synchronous Reluctance Rotors With Multi-Material Additive Manufacturing

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Abstract—Synchronous reluctance (SynR) machines are promising rare-earth material-free alternatives to permanent magnet machines. However, structural challenges limit their operating speed and power density. This paper proposes and investigates multi-material additive manufacturing (MMAM) as a key-enabler to realize power-dense and high-speed SynR machines. It does so by proposing designs that guide magnetic flux through solid rotors realized by selective placement of magnetic and non-magnetic materials. To explore this concept, first, material samples are additively manufactured and experimentally characterized to assess the structural and magnetic properties that can be expected for the proposed rotors. Second, the design space of each rotor type is explored using these measured properties within finite element analysis. The results reveal that MMAM can enable fabrication of SynR motors with power density levels that are at the leading edge of all conventional electric machine topologies. It is shown that tip speeds in excess of 300 m/s can be achieved, resulting in 3-4x improvement in power density over conventional SynR motors. A solid SynR rotor is printed in an experimental MMAM laser powder bed fusion system. The rotor is paired with an existing stator to create a functional SynR motor with a saliency ratio of 2.59 and torque rating of 4.15 Nm. This is the first publication of a SynR rotor prototype constructed via MMAM.

Index Terms—Synchronous reluctance, additive manufacturing, multi-material additive manufacturing, 3D printed electric machines, optimization, power density, rare-earth-free machines.

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I. INTRODUCTION

WITH the advent of electrification, requirements towards improved energy use have demanded cleaner, more efficient electric machines. Synchronous reluctance (SynR) machines provide several benefits over their counterparts. This includes a lack of environmentally-harmful and expensive rare-earth permanent magnet (PM) material compared to PM machines, and higher torque density and efficiency compared to induction machines [1]. However, SynR machines have limited speed and power density capabilities due to complex rotor geometries, which result in reduced mechanical strength at high operating speeds [2]. Improving the structural properties of SynR rotors can yield SynR machines with increased power density, positioning them as solutions to realize power-dense and rare-earth-free electric machine designs.

This paper proposes to enhance the mechanical strength of SynR machines by altering air flux barriers within the rotor to instead be made of a strong, solid material. Additive manufacturing (AM) provides an avenue towards doing so and has been applied to the fabrication of electric machine components in different capacities, e.g.: [3], [4], resulting in improved machine performance. Multi-material additive manufacturing (MMAM) is a newly-developed technology within AM that utilizes multiple materials across a single geometric body [5]. This has a primary advantage that enables a single feature to contain different material properties across a continuous region. Its application could unlock new SynR designs absent of the structural issues hindering conventional designs. This would allow SynR machines to operate at higher speeds and become more power-dense.

The conference version of this paper [6] was the first work to introduce MMAM technology and model the performance potential of using MMAM to realize solid high-speed SynR rotors. This paper furthers the investigation in [6] and fabricates and characterizes a solid SynR rotor design using dual-material techniques, with Fe-3.5Si as the magnetic material. Recently, the authors of [7] followed this approach to develop a prototype of a solid SynR rotor using 17-4PH steel as its magnetic material. This material has inferior magnetic properties compared to the Fe-3.5Si [8] used in this paper, but more readily fuses with stainless steel.

The key contributions of this paper are i) the proposal of a MMAM SynR rotor, ii) an exploration into how the design space of SynR motors can be transformed by using

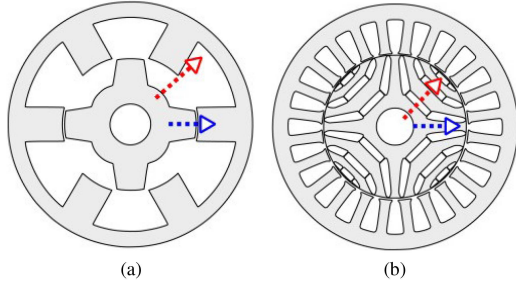


Fig. 1. Rotors with (a) smaller and (b) larger saliency ratios [15].

MMAM techniques, and iii) experimental validation of the manufacturability and performance of the first MMAM SynR design to be published. This paper initially reviews SynR electric machine technology (Section II). Multi-material additive manufacturing technology is then introduced, its application to SynR machines is discussed (Section III), and materials to realize solid high-speed SynR rotors are selected and experimentally characterized (Section IV). An optimization framework is then developed to explore the design space for solid SynR rotors enabled by MMAM in Section V. Results are presented and selected designs are identified and analyzed in Section VI. Finally, a prototype design is selected, printed, and experimentally characterized in Section VII.

II. SYNCHRONOUS RELUCTANCE TECHNOLOGY

SynR machine design aspects relevant to this paper are now reviewed. Important characteristics considered include saliency ratios, which reflect torque, and machine speed and power capabilities. Considering all of these characteristics is necessary to identify the critical design aspects to achieve high-quality SynR rotor designs using MMAM.

A. Operating Principle

Speed and power density limits of conventional SynR machines result from their complex rotor geometries that utilize small or thin geometric features. These features are required to maximize torque, but lead to high induced rotor stresses. Reluctance torque T of any electric machine is related to the number of pole pairs p , d- and q-axis currents $i_{d,q}$, and d- and q-axis inductances $L_{d,q}$ [9], per (1).

$$T \propto p(L_d - L_q)i_d i_q \quad (1)$$

Reluctance torque is generated by employing a non-uniform rotor shape to create differences in the d- and q-axis inductances. To maximize reluctance torque, the difference between L_d and L_q must be maximized. In conventional SynR design, that leads to a complex and structurally-weak rotor. To detail this principle, Fig. 1 shows two rotors with different saliency ratios (L_d/L_q). Fig. 1(a), has a lower saliency ratio than Fig. 1(b), and if excited by the same current will produce less torque. Conversely, Fig. 1(a) contains a stronger rotor geometry than Fig. 1(b), and thus is able to operate at higher speeds.

B. Machine Output Quantifications

Machine output power P is a key quantification because it directly relates to power density, which should be maximized. Additionally, the speed-power metric $\omega\sqrt{P}$ indicates the maximum power that a machine can reasonably be designed for given a maximum operating speed [10], [11]. Expressions for power and speed-power are derived from geometric and machine variables and can be evolved to:

$$P = T\omega = 2V_r\sigma_{F,\tan}\omega = 2\pi rL\sigma_{F,\tan}v_{\text{tip}} \quad (2)$$

$$\omega\sqrt{P} = \sqrt{2\pi\sigma_{F,\tan}v_{\text{tip}}^3\left(\frac{L}{r}\right)} \quad (3)$$

in terms of the machine parameters of rotor volume V_r , rotational speed ω , tip speed v_{tip} , rotor radius r , stack length L , and magnetic shear $\sigma_{F,\tan}$. Tip speed is an indication of how fast the edge of the rotor can travel, and is commonly used to quantify the structural and electrical performance of electric machines. Synchronous reluctance machines have maximum tip speeds of 130–170 m/s. This is lower than that of laminated, non-salient and induction machines that can achieve tip speeds of 200–250 m/s [12]. The tip speed limitation in SynR machines restricts their usefulness because this limits power and speed-power as seen in (2) and (3).

Given their tip speed restrictions, SynR machines are often used for slower-speed applications (4,000–6,000 RPM). Typical efficiencies for SynR machines are >95% [13]. The highest performance SynR machines occur at high slot/pole ratios, with the process described in [14] being the design method for the conventional SynR rotor construction used in this study. Power (and thus power density) and the speed-power metric could be increased through innovation in the rotor design to increase structural integrity. This paper investigates how new MMAM techniques can be used to enhance SynR machine performance by mitigating sources of stress concentrations that restrict conventional designs.

III. MULTI-MATERIAL ADDITIVE MANUFACTURING

This section reviews the application of MMAM to SynR technology and describes the printing process used in this paper. While several challenges of SynR rotor design can be addressed by using MMAM, this work focuses on overcoming restrictions posed by conventional flux barrier design. This section explores capabilities and limitations of MMAM and their impact on the proposed design.

A. Application to Synchronous Reluctance Machines

Advancements in MMAM have seen an increase in recent years, and the field of electric machines contains many possibilities for its implementation. Existing literature proposes the replacement of air flux barriers in conventional SynR machines with a metallic insulator [3]. In this scenario, stainless steel with magnetic permeability properties comparable to air, is proposed as an ideal replacement.

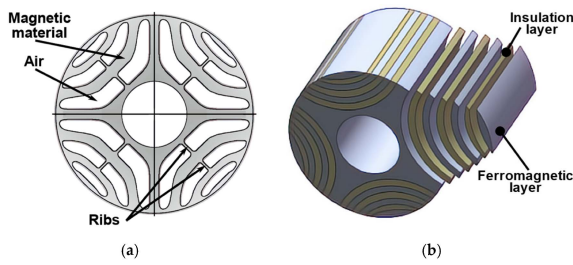


Fig. 2. Design of synchronous reluctance rotors [3].

The removal of the ribs in SynR rotors increases the power density and efficiency [13], [16], and improves the structural integrity of the rotor. The load that would act on the structural ribs is now spread out across the entirety of the magnetic flux barrier/flux carrier interface, drastically reducing the stress concentrations within the rotor. This concept is illustrated in both Fig. 2(a), which depicts a conventional SynR rotor, and Fig. 2(b), which diagrams a conceptual transversely-laminated rotor design with magnetic and insulation material layers. In this new design, maximum rotor speed and power density can be increased, therefore removing the two main limitations of SynR machines. New MMAM technology can be used to fabricate this design since the process can form a metallic bond between the two materials and seamlessly integrate stainless steel flux barriers into the rotor design.

B. Physical Implementation and Restriction

The two main challenges of implementing MMAM in the field of electric machines are limited availability of commercial MMAM machines capable of handling at least two materials simultaneously and limited knowledge of bonding mechanisms of dissimilar material within MMAM context [17]. The AM process used in this work is laser powder bed fusion (LPBF). This method was chosen because it is the most widely used metal additive manufacturing method with the highest geometric resolution [5]. To create the solid structure via LPBF, a laser beam is used to deliver thermal energy to a powder bed in order to selectively melt metal powder which rapidly solidifies to create 3D geometries in an additive, layer-by-layer fashion [18].

The authors have modified an existing LPBF system (EOS M290) to incorporate MMAM capability. The modified system has proven capable of creating metallic bonds between dissimilar materials [19]. The EOS M290 is a single material printer that uses a powder reservoir and a recoater blade to distribute new powder onto the build plate for each layer. To achieve multi-material capability, the reservoir was sub-divided using thin removable dividers while loading, to accommodate multiple metal powders side by side. While this can limit the print capabilities since the powders can mix when the dividers are pulled out during the recoating process, it is deemed acceptable for this proof of concept process. The process is detailed in Fig. 3(a)–(e). Fig. 4(a) details the layout of the multi-material LPBF approach used in this work. This system is only capable of creating two dimensional interfaces in vertical or horizontal directions with respect to its build direction. This restriction is the limiting factor

in the proposed MMAM rotor designs. Future advancements in multi-material LPBF systems will soon lift these restrictions and allow for more complex MMAM rotor designs [20].

IV. MATERIAL CHARACTERIZATION

This section details the decision making processes to select each MMAM material so that it may be printed and characterized prior to the design and MMAM fabrication of the SynR rotor. Both the magnetic material and flux barrier material are chosen based on their respective properties and function to optimize the performance of the MMAM SynR machine. Once the materials are chosen, each is independently printed, using the aforementioned LPBF process, and characterized. These characterization results are fed to FEA models documented later in the paper to investigate MMAM rotor concepts.

A. Material Selections

Magnetic material constitutes the bulk of the rotor and should have ideal properties for use in electric machines. High permeability, high saturation, and high thermal conductivity are desired [21]. Fe-Si with 3.5 weight % Si (Fe-3.5Si) is chosen as the magnetic material for this study because it has proven to be manufacturable using LPBF and commonly used for electric machine cores. Although increasing the Si content beyond 3.5% can limit eddy current losses, studies have shown that Fe-Si with high silicon content is highly brittle and therefore difficult to manufacture [3], [21]. As such, Fe-3.5Si is the most practical both for implementation in an electric machine and as a printable material via LPBF [22]. The material making up the flux barrier should have permeability similar to that of air. It should also possess mechanical properties similar to that of Fe-3.5Si for adequate structural integrity. Furthermore, the flux barrier material should form a metallic bond with Fe-3.5Si that is free of inter-metallic compounds and defects during MMAM processing. Stainless steel 316L fits the above criteria and was therefore chosen as the material that will make up the flux barriers.

B. Experimental Validation of MMAM Capability

The key manufacturing parameters in LPBF are laser power, laser speed, hatch spacing, and layer thickness. These parameters are optimized to avoid the formation of defects, i.e. porosity or cracks, during LPBF. Defects can adversely affect mechanical and magnetic properties of the print [18]. The process parameter optimization for Fe-3.5Si and Fe-3.5Si-316L multi-material that has been carried out is now described.

1) *Mechanical Properties:* The maximum speed of MMAM rotors is defined as the speed at which any part of the rotor reaches the material yield strength. The main stress that rotors experience during operation occurs when centrifugal forces put the rotor in tension, concentrated mostly on the outer rotor surface. Because of this, focus is placed on tensile testing. First, the MMAM process is refined through iterative printing and testing of single-material samples. The test procedures outlined in ASTM Standard EM/EM8 [23] are used to determine the

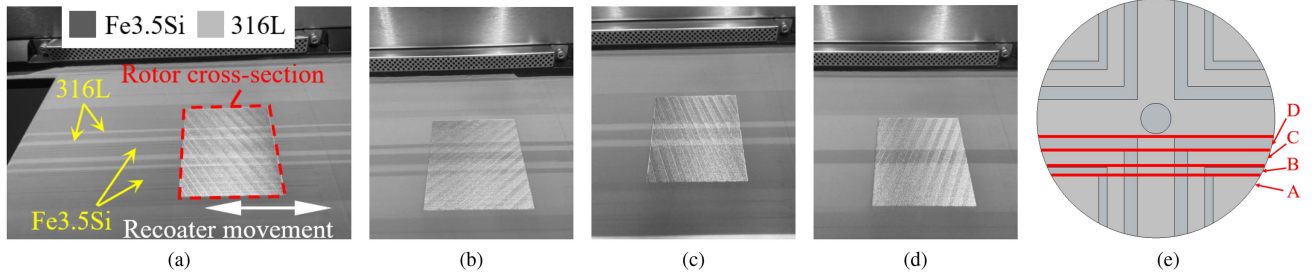


Fig. 3. (a) Photo and labeling of multi-material print layer for section A of rotor geometry; (b)-(d) Photos of print layers for sections B, C, and D of rotor geometry; (e) Diagram of sections A - D of rotor geometry.

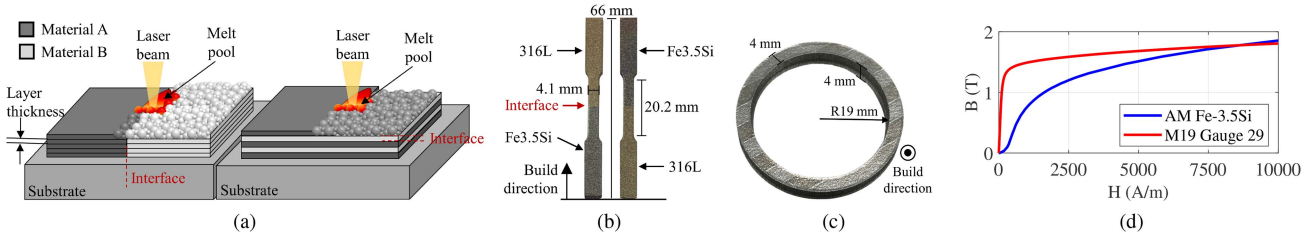


Fig. 4. (a) Layout of the multi-material LPBF method used in this work for printing vertical and horizontal material interfaces; (b) Multi-material Fe-3.5Si-316L tensile and (c) Fe-3.5Si ring test specimens; (d) B-H curve for the Fe-3.5Si compared to a standard magnetic steel (M19).

TABLE I
EXPERIMENTALLY MEASURED STRUCTURAL PROPERTIES

Mechanical Property	Fe-3.5Si	316L	Units
Young's Modulus	119	140	GPa
Yield Strength	495	460	MPa
Ultimate Strength	561	513	MPa

mechanical properties of each printed material. Table I shows the final results after process refinement.

Second, the bond strength between the two materials is investigated to produce an MMAM process that reliably creates a multi-material bond that is stronger in tension than the material with the lower yield strength. Multi-material tensile specimens were fabricated (see Fig. 4(b)) and tested. Details of the test procedure can be found in [19]. Results from the final process show that necking and fracture does not occur at the interface of the two materials, proving the interfacial bond does not yield before the material with the lower yield strength. This means that the highest stress throughout the multi-material rotors will occur in the 316L stainless steel material, not along the interfaces between the Fe-3.5Si and 316L, and that MMAM rotors are not compromised by Fe-3.5Si-316L bond strength.

2) *Magnetic Properties:* Single material ring samples were printed and tested to determine the B-H curve of printed Fe-3.5Si. The samples were heat treated at 1000C for 1 hour and naturally cooled in vacuum conditions and tested according to ASTM Standard A773 [24]. The measured B-H curve is shown in Fig. 4(d) and shows that the magnetic properties of printed Fe-3.5Si strongly resemble the popular electric steel M19 29 Gauge, providing confidence it can be used in SynR rotors. The results from both the tensile and ring tests are used in FEA

models presented later, in Section VI, to explore the design space for the proposed MMAM rotor.

V. MULTI-PHYSICS OPTIMIZATION FRAMEWORK

This section formulates an optimization problem and develops a multi-physics modeling framework to investigate the potential of MMAM to improve SynR machine performance. Four distinct rotor topologies are evaluated using this framework, first with fixed speed (study I), and next with variable speed (study II). The goals of this framework are to quantify and compare the design space of SynR machines manufactured using conventional methods against machines manufactured with MMAM technology, with and without the printing restrictions of today's MMAM equipment.

A. Optimization Problem and Modeling Framework

The optimization framework performs structural and electromagnetic analyses of each machine design. The four SynR rotor topologies are shown in Fig. 5 and include i) a baseline design ("Conventional") composed of standard M19 29-gauge steel laminations and three designs assumed to be printed with MMAM: ii) a V-shaped flux barrier design ("V"), iii) a square-shaped flux barrier design ("Square") that were conceptualized to account for the restrictions of today's MMAM equipment discussed in Section III-B and iv) a "Vision" design that would be possible if today's MMAM constraints are overcome.

The same framework is used with minor differences for two different studies, as shown in Fig. 6. The structural and electromagnetic analysis steps utilize 2D FEA in the JMAG software package. The conventional rotor is modeled using the standard B-H curve, Young's modulus (185 GPa), and yield strength

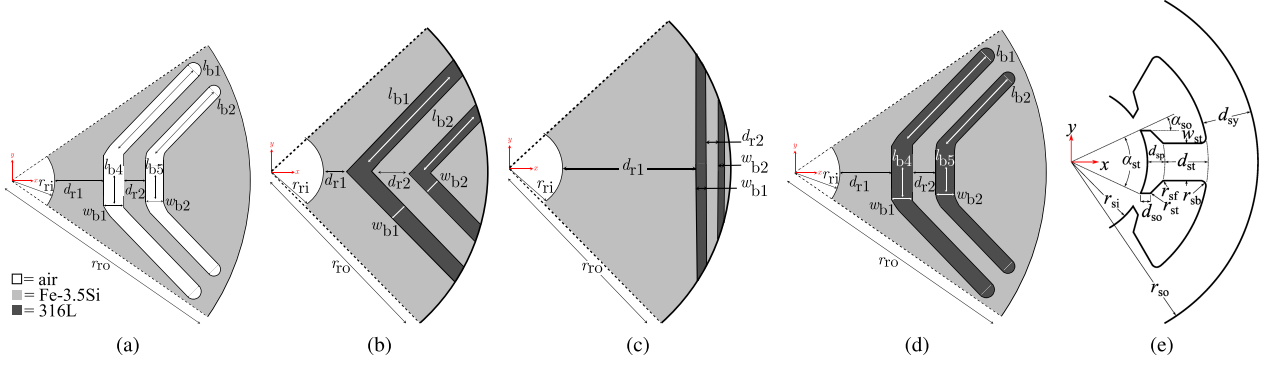


Fig. 5. Parameterized cross-sections of: (a) conventional rotor, (b) V-shaped rotor, (c) square rotor, (d) vision rotor, and (e) stator.

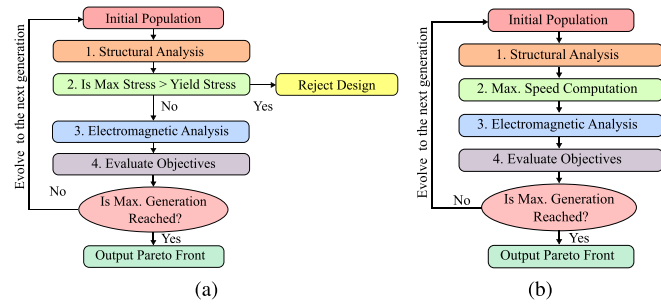


Fig. 6. Flow charts for design space exploration study at (a) fixed rotor speed and (b) variable rotor speed.

TABLE II
ROTOR VARIABLES AND STUDY II SELECTED DESIGNS

	Range	Conv.	V	Square	Vision	Units
d_{r1}	[3.2, 16]	3.692	4.799	27.8	3.845	mm
d_{r2}	[4.8, 16]	6.441	9.931	3.930	6.471	mm
l_{b1}	[20, 30]	29.98	-	-	29.97	mm
l_{b2}	[21.6, 32.4]	21.67	-	-	21.63	mm
l_{b4}	[9.6, 14.4]	13.60	-	-	14.28	mm
l_{b5}	[4.8, 7.2]	6.763	-	-	6.594	mm
r_{ri}	[3, 12]	11.46	6.348	9.950	11.31	mm
r_{ro}	[45, 50]	49.98	49.99	49.99	49.99	mm
w_{b1}	[2, 8]	4.763	5.543	2.660	4.783	mm
w_{b2}	[2, 8]	4.669	3.609	3.942	4.697	mm
ω_{frac}^1	[60, 100]	98.4	65.5	64.0	68.8	%

¹ ω_{frac} only used in study II.

(450 MPa) for M19 steel. The MMAM designs use the measured Fe-3.5Si B-H curve shown in Fig. 4(d) and the measured Young's modulus and yield strength values reported in Table I. All electromagnetic analysis are transient and evaluated over a single electrical period. Loss calculations include iron losses (hysteresis and eddy current) and DC conduction losses in the stator conductors. While AC losses (including proximity effects) in the stator conductors can be significant at higher operating speeds, these can be mitigated by wire choice (i.e., using Litz wire). Since this work is primarily focused on evaluating the performance gains by using MMAM rotors, AC conductor losses are not considered. The rotor structural integrity is assessed by determining whether the maximum stress exceeds the yield stress, also known as the distortion energy theory method [25]. Stresses in this framework are found using the von Mises stress model, considering a safety factor of 1.5x. Since this study is concerned with high rotational speeds, it is important to model windage losses on the rotor. Analytic expressions from [26] are used for this purpose.

To search the design space, the multi-objective evolutionary algorithm based on decomposition (MOEA/D) [27] is used in conjunction with the modeling framework through open-source Python scripts [28]. The optimization objectives are defined in (4), where $P_{density}$ is the volumetric power density, P_{mech} is the mechanical power output, $V_{machine}$ is the volume of the machine, r_{so} is the outer stator radius, T_{avg} is the average shaft torque, ω is the rotational speed, L is the axial length, η is the efficiency, P_{loss} is the total losses in the machine, T_{rip} is the torque ripple,

and T_{max} and T_{min} are the maximum and minimum values of torque over one cycle.

$$\begin{aligned}
 O_1 &= \begin{cases} -P = -T_{avg} \times \omega & \text{for Study I} \\ -P_{density} = -\frac{P_{mech}}{V_{machine}} = -\frac{T_{avg} \times \omega}{\pi r_{so}^2 L} & \text{for Study II} \end{cases} \\
 O_2 &= -\eta = -\frac{P_{mech}}{P_{mech} + P_{loss}} \\
 O_3 &= T_{rip} = \frac{T_{max} - T_{min}}{T_{avg}} \quad (4)
 \end{aligned}$$

The optimization free variables are the geometric parameters of the SynR machine's radial cross-section, with rotor parameters defined in Fig. 5(a)–(d) and stator parameters defined in Fig. 5(e). A $Q = 36$ slot, $p = 2$ pole-pair stator is used in each study. All machine designs are modeled with a fixed axial length of 70.61 mm, corresponding to an existing stator that is used for prototype fabrication (described in Section VII-A).

B. Study I: Fixed Speed

The first study investigates the four rotor designs at a fixed rotational speed of 20,000 RPM so that each machine has the same electric frequency. This study allows designs to achieve their structural limit by varying the rotor diameter and thus provides insight into the relationships between power and efficiency capabilities. The rotor and stator parameter ranges are listed in Tables II and III. In each design, the total stator slot current

TABLE III
STATOR VARIABLES

	Study I Range	Study II Value	Units
α_{so}	[3.75]	3.75	deg
α_{st}	[7.5]	7.5	deg
d_{so}	[0.25, 8]	1	mm
d_{sp}	[0.5, 16]	2	mm
d_{st}	[10, 120]	16.52	mm
d_{sy}	[5, 100]	9.39	mm
w_{st}	[2, 20]	3.25	mm
Q	[36]	36	slots
r_{sb}	[0.1]	0.1	mm
r_{sf}	[0.1]	0.1	mm
r_{si}	[45.5, 52.5]	50.5	mm

is determined from the cross-sectional area of the stator slots by assuming a slot-fill factor of 0.5 and a current density of $4 \text{ A}_{\text{rms}}/\text{mm}^2$.

The structural and electromagnetic analyses steps in Fig. 6(a) are both evaluated at a rated speed of 20,000 RPM. The electromagnetic analysis is conducted at rated torque.

C. Study II: Variable Speed

This second study investigates the potential to realize high-speed and power dense SynR designs using MMAM. This is done by holding the stator design constant and allowing the rated rotor speed to vary. The stator design corresponds to an existing stator that will be used to test the prototype rotor (see Section VII-B). The free variables and their ranges are listed in Table II and the fixed stator dimensions are listed Table III.

The following series of steps in Fig. 6(b) are used to determine the rotor's rated speed ω_{rated} :

1) *Structural Analysis*: The design is analyzed to compute its maximum stress at four different shaft speeds. These speeds are determined as 1/4, 1/2, 3/4, and 1 times a hypothetical maximum speed for each rotor type. This theoretical maximum speed is taken as 20,000 RPM for the conventional rotor and 60,000 RPM for the V-shaped, square, and vision rotors, based on the yield strength of each material and maximum conventional tip speeds [3].

2) *Max. Speed Computation*: The maximum stress at each speed from step 1 is curve fit to a second order polynomial function of speed. The maximum allowable speed is determined by interpolating the polynomial to find the speed ω_{max} where the maximum stress has a 1.5x safety factor as compared to the yield stress.

3) *Compute Rated Speed*: This design study introduces an additional free variable ω_{frac} that is used to determine the rotor's rated speed by derating the maximum speed, as $\omega_{\text{rated}} = \omega_{\text{frac}} \times \omega_{\text{max}}$. This is done to allow the optimization algorithm to consider designs that operate below the maximum achievable tip speed of each rotor.

VI. MULTI-PHYSICS OPTIMIZATION RESULTS

This section presents results from the design space exploration of the conventional and MMAM SynR machine design variants using the multi-physics optimization framework developed in

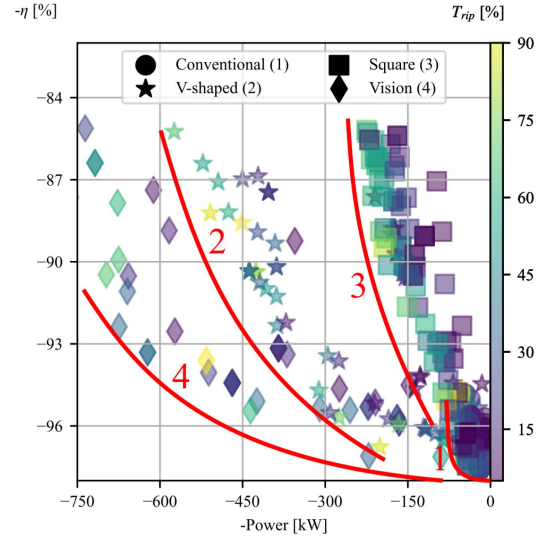


Fig. 7. Design study I Pareto fronts at 20,000 RPM and rated torque.

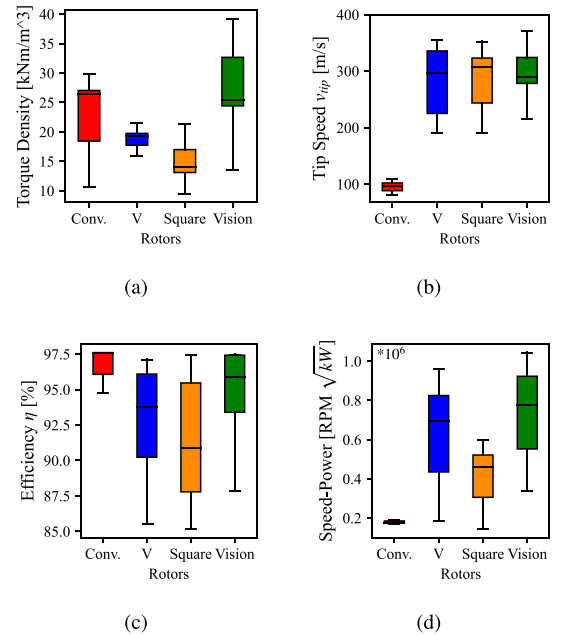


Fig. 8. Design study I: statistical distribution of performance metrics for designs on Pareto front of each rotor type.

Section V to understand the general design trends between the rotor variants and to draw conclusions on the potential impact of MMAM on SynR machine design.

A. Study I Results: Fixed Speed

The Pareto dominant designs from the fixed speed study are depicted in Figs. 7 and 8. From Fig. 7, it is evident that all MMAM rotor topologies are able to achieve significantly higher power ratings for a given efficiency, with the vision rotor reaching a $> 7\times$ power increase, as compared to the conventional rotor design. Insights into how this power improvement

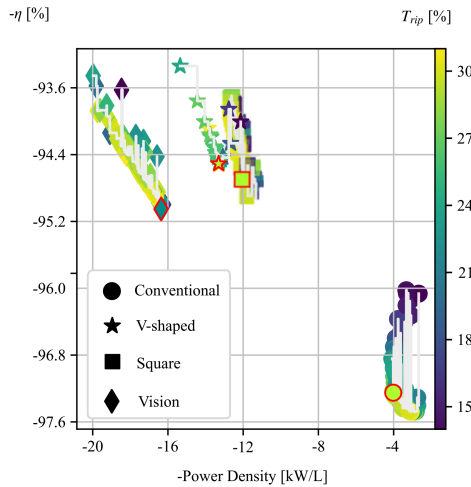


Fig. 9. Design study II Pareto fronts at rated speed and torque.

is achieved are revealed in Fig. 8(a) and (b). All rotors have a similar torque density (with the V-shaped and square rotor designs actually performing lower than the conventional rotor), but the MMAM rotors are able to achieve an impressive increase in tip speed. All MMAM rotors are able to reach tip speeds over 300m/s, which is over twice the upper limit identified for the strongest SynR designs in Section II-B, $> 3\times$ the tip speed of the conventional rotors in Fig. 8(b), and amongst the highest tip speeds reported in literature for any motor type [11].

The MMAM rotors' high tip speeds are consistent with the discussion in Section III-A on replacing air flux barriers with printed stainless steel to produce a single, solid rotor that eliminates stress concentrations. The implications of increases in tip speed were shown in (2) and (3) to result in large increases in power and speed-power capability. Fig. 8(d) confirms a nearly $5\times$ improvement in speed-power capability.

Figs. 7 and 8 also reveal performance differences amongst the MMAM rotors. Since the vision rotor is free from the restrictions of today's MMAM technology (discussed in Section III-B), it has the most degrees of freedom and correspondingly achieves the highest power, torque density, efficiency, and speed-power. The V-shaped and square rotor designs perform lower on all metrics, with the V-shaped rotor performing the best (capable of approximately twice the power output of the square rotor), but are still able to reach significantly higher power and speed-power than the conventional rotor.

B. Study II Results: Variable Speed

The resulting Pareto dominant designs from the variable rotor speed study are shown in Figs. 9 and 10. By using a fixed stator design and allowing only the rotor speed to vary, the study aims to explore differences in power density and identify a rotor to print using MMAM. Fig. 10(a) shows that the MMAM rotors are able to achieve 3-4 $\times$ higher power density and Fig. 10(b) shows a similar increase in rated speed (reflecting the MMAM rotor's ability to support higher tip speeds). Interestingly, Fig. 10(d) reveals that the vision rotor is able to achieve 33% higher magnetic shear than the conventional rotor. This is attributed

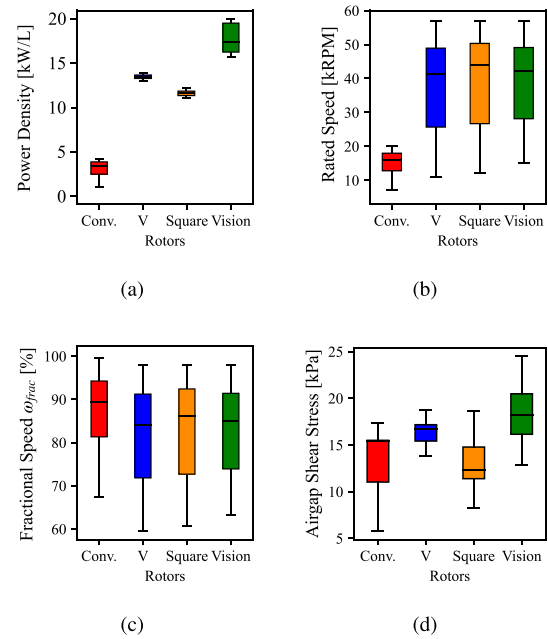


Fig. 10. Design study II: statistical distribution of parameters for designs on Pareto front of each rotor type.

TABLE IV
PERFORMANCE OF SELECTED DESIGNS FOR DESIGN STUDY II

Parameter	Conv.	V	Square	Vision	Units
Power Density	4.0	13.3	12.1	16.4	kW/L
Torque Ripple	31.6	27.7	26.6	21.8	%
Efficiency	97.5	97.3	97.4	97.5	%
Maximum Speed	20,244	57,473	57,207	57,414	RPM
Tip Speed	106	301	300	301	m/s
Saliency Ratio	5.91	4.48	4.26	5.95	-
Airgap Shear Stress	17.6	19.9	19.2	24.8	kPa

to the lack of stress concentrations in the vision rotor allowing for better optimization of the electromagnetic performance.

Fig. 9 reveals that the power density is in trade-off with efficiency. This is because increases in power density are primarily achieved through increases in rotational speed and thereby the electric frequency, resulting in more iron losses. This is particularly true in the stator, which is modeled as M19 29 Gauge steel. The highest power density designs have rated speeds above 40,000 RPM (>1.3 kHz electric frequency). Practical electric machine designs in this frequency range use ultra-thin gauge stator steel to decrease stator losses, meaning that the reduced efficiency observed here is not a trend inherent to the power-dense MMAM SynR rotors. Note that this is reflected in ω_{frac} in Fig. 10(c): the MMAM rotors include lower values of ω_{frac} so that the rotor's rated speed is lower than the maximum speed that the rotor structure can withstand.

Within the MMAM rotors, the vision rotor again significantly outperforms the two rotor structures that are printable with today's MMAM equipment. Of these two printable rotor structures, the V-shaped rotor achieves modestly higher power density ratings than the square rotor structure, making it a promising design topology for prototype fabrication.

TABLE V
COMPARISON OF RESULTS OF SYN Rotor STUDIES TO CONVENTIONAL ELECTRIC MACHINES

Parameter	Conv.	V	Square	Vision	Conventional SynR	Surface PM	Induction	Units
Power Density	4.0	15.8	12.1	20.0	≤ 7	≤ 10	≤ 6	kW/L
Speed-Power Metric	6.9×10^4	8.9×10^5	4.6×10^5	1.0×10^6	$\leq 3.4 \times 10^5$	$\leq 8 \times 10^5$	$\leq 6 \times 10^5$	RPM $\sqrt{\text{kW}}$
Saliency Ratio	5.91	4.48	4.26	5.95	≤ 8	1	1	-
Maximum Tip Speed	106	379	378	386	130-170	200-250	150-200	m/s
Airgap Shear Stress	17.6	19.9	19.2	24.8	≤ 33.5	≤ 36.0	≤ 21.5	kPa

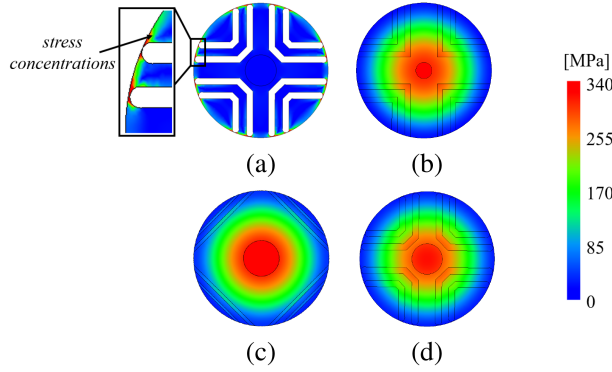


Fig. 11. Stress distributions for selected designs: (a) conventional, (b) V-shaped, (c) square, and (d) vision rotor designs at each design's rated speed.

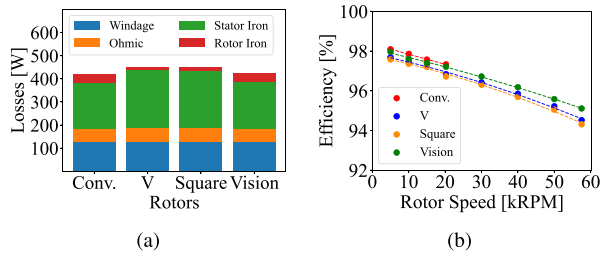


Fig. 12. Selected designs: (a) losses at 20,244 RPM and (b) efficiency.

C. Detailed Design Analysis and Discussion

Differences in high performing designs of each rotor type are now analyzed. First, a promising design of each rotor is selected and analyzed. Second, the capabilities of each rotor are compared against conventional machines in Table V.

The designs were selected on a basis of maximizing power density while obtaining reasonable efficiency and are shown as red outlines in Fig. 9. Their dimensions are listed in Table II and performance parameters in Table IV. The design cross-sections are overlaid with stress distributions in Fig. 11 at their maximum operating speed. As expected, stress concentrations in the conventional design are significant between the flux barriers and outer edge of the rotor, limiting its maximum operating speed. The addition of the solid flux barrier significantly reduces the stress concentrations. The V-shaped, square, and vision rotor designs exhibit comparable maximum speeds since their geometry is the same solid metal structure but configured with different arrangements of Fe-3.5Si and 316L.

The loss mechanisms for each rotor are shown in Fig. 12(a). Since several loss components are speed dependent, each selected design is re-analyzed at the same speed (conventional design speed) to facilitate accurate comparisons. The difference in iron losses is notable between the V-shaped/square and conventional/vision designs. The stator iron losses appear higher in the V-shaped/square rotors but rotor iron losses appear lower. This is attributed to these rotors bringing the flux barriers directly to the outer rotor surface, and thereby increasing the higher order rotating field harmonics that the stator teeth are exposed to.

Machine efficiencies (O_2 in (4)) are calculated using the resulting power and losses from each design state to compare rotor performances up to each design's maximum speed. The resulting efficiency profiles are shown in Fig. 12(b). The conventional and vision rotors exhibit similar efficiency curves, which is expected as their magnetic cross-sections are very similar. The vision rotor can operate much faster where it only then becomes less efficient. The V-shaped and square rotors exhibit a slightly lower efficiency profile due to their lower saliency ratios and torque output.

Finally, Table V compares the upper limits calculated for the four SynR rotor topologies with conventional electric machine types. Data for the conventional machines was assembled from [11], [12], [29], [30], [31], while data of the four SynR rotor types is taken as the upper quartile value of each metric from designs in the Pareto Fronts of Figs. 7 and 9. The V-shaped, square, and vision rotors are shown to exceed the maximum tip speed of all other machines through their solid-cylindrical cross-section. Even with today's printing restrictions, the implementation of MMAM in SynR rotors allows them to exceed the typical power density and speed-power ranges of conventional SynR, surface PM, and induction machines. This confirms that MMAM has potential to enable SynR motors to be amongst the highest power density motor technologies.

VII. EXPERIMENTAL VALIDATION

This section evaluates the feasibility of using MMAM to print a SynR rotor and validates the FEA models used by Section VI to explore the MMAM rotor design space. The V-shaped rotor design presented in Section VI-C is selected to print due to its superior performance relative to other printable designs. The section presents the prototype fabrication and challenges that were encountered and prototype test results.

A. MMAM of Prototype Rotor

The MMAM process for Fe-3.5Si and 316L presented in Section III was used to fabricate the rotor inside the authors'

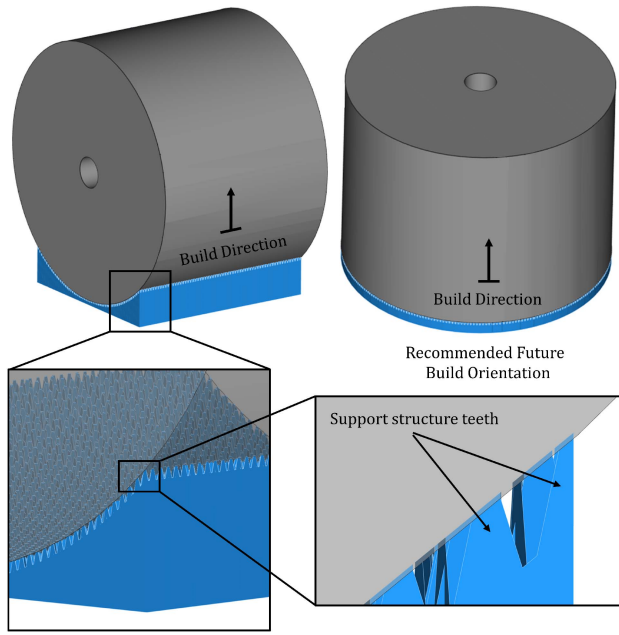


Fig. 13. Supporting structure and build direction.

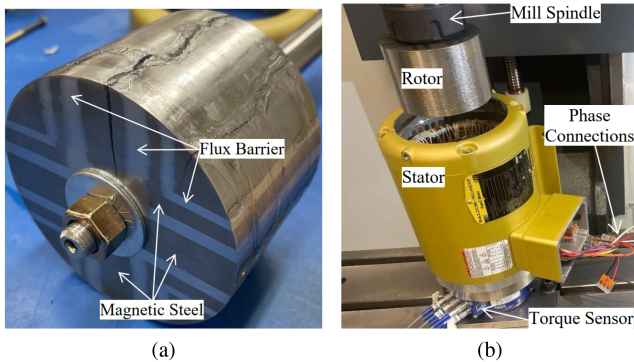


Fig. 14. (a) Fabricated rotor; (b) Experimental setup.

modified LPBF system (see Figs. 3 and 4(a)). The design dimensions are listed in the “V” column of Table II. The design was extended to be printed as an oversized cylinder and machined down to a 100 mm diameter to ensure tight tolerances within the airgap.

The initial manufacturing attempt resulted in severe cracking that made it impossible to complete the print. The authors determined this to be due to the mismatch in coefficients of thermal expansion of Fe-3.5Si and 316L creating residual stresses in the rotor. This is exacerbated by the repeated thermal cycling inherent to the LPBF process and the high brittleness of Fe-3.5Si compared to the ductile 316L. To mitigate this issue, a 316L support structure was designed for a subsequent print. The design is shown in Fig. 13 and consists of a 1x1 mm grid of thin walls connected to the rotor with a trapezoidal feature known as *teeth*. These teeth allow the rotor to experience strain under thermal stress and increase the heat transfer rate to the base plate, reducing induced residual stress.

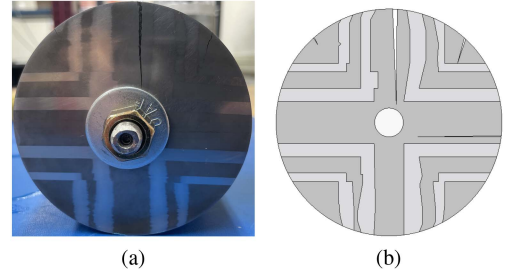


Fig. 15. (a) Printed rotor with non-idealities. (b) FEA model of (a).

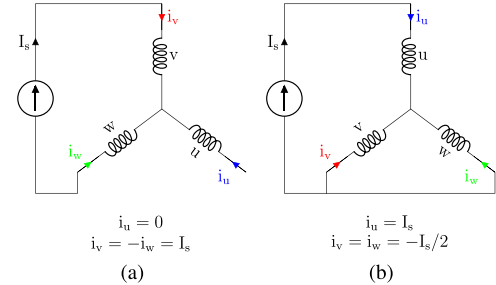


Fig. 16. Circuit diagrams for d- (a) and q- (b) axis excitation.

TABLE VI
INDUCTANCE AND SALIENCY RATIO EXPERIMENTAL RESULTS

Variable	FEA: Design	FEA: Printed	Experimental
L_d [mH]	76.0	40.1	48.9
L_q [mH]	16.9	18.0	18.9
$L_d - L_q$ [mH]	59.1	22.0	30.0
Saliency ratio	4.48	2.23	2.59

Incorporating the support structure made it possible to complete the print; however, it was not sufficient to fully prevent cracking, as the final multi-material rotor exhibited cracks in the brittle Fe-3.5Si sections. These cracks were filled with cyanoacrylate [32] before machining to allow for movement and prevent further propagation. The resulting rotor assembly is shown in Fig. 14(a). The Fe-3.5Si and 316L regions are both clearly visible and the severity of the cracks can be observed in the radial cross-section and along the axial length. In addition to cracking, it is evident that portions of the flux barriers became misaligned during the print.

Both of these deficiencies (cracking and flux barrier misalignment) are artifacts of the authors' experimental procedure and modified LPBF system and will be overcome in future professional-grade MMAM systems. For example, to prevent cracking, the build direction can be modified from radial to axial (right side of Fig. 13) to reduce the residual stresses experienced by the rotor. This stress reduction is because the axial build direction reduces the mismatch surface area at any given cross-section. However, the axial build direction is not possible in the authors' modified LPBF system. Other approaches to eliminate cracking include alloying solutions to increase the ductility of Fe-3.5Si and optimizing the scan strategies to minimize residual stress, all of which would be readily accomplished in a dedicated MMAM system. Likewise, the misalignment within the

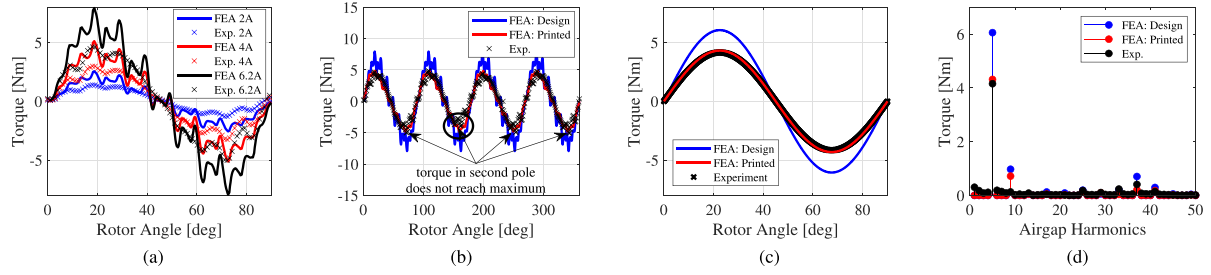


Fig. 17. Comparison of experimental and FEA torque profiles: (a) comparison with FEA: Design for one electrical cycle, (b) for one mechanical cycle, (c) fundamental harmonic only at rated current, and (d) spectral components at rated current.

flux barriers arises from the ad hoc method of multi-material deposition employed in the modified LPBF machine. However, multi-material powder deposition modules with resolutions as fine as 300 μm are now becoming commercially available [33]. These units enable precise and controlled placement of different powders on the powder bed, ensuring an exact match between the CAD model and the final printed component.

B. Test Setup

The printed rotor was tested with a commercial NEMA 145T stator shown in Fig. 14(b), repurposed from an induction motor. The stator dimensions are listed in Table III and were used in the design space study of Section VI-B. To characterize the prototype machine, a CNC mill is configured as a dynamometer. The rotor is installed in the mill spindle and the stator is mounted on a load cell fixed to the mill table. The load cell allows measuring the reaction torque on the stator. The complete test setup is shown in Fig. 14(b).

C. Test Procedure and Experimental Results

The objectives of the tests are to validate the modeling approach used in the design studies and to demonstrate the MMAM process capability for fabricating functional electric machine components. The deformations that resulted from thermal cycling during the print process alter the expected magnetic behavior from what was presented in Section VI. To account for this, the FEA model was modified to use the structure shown in Fig. 15(b). This cross-section was constructed by measuring the lengths of the cracks, their locations on the rotor, and the dimensions of the misaligned flux barriers. In the following material, calculations that were obtained with this revised model are labeled as “FEA: Printed;” calculations for the rotor as-designed are labeled as “FEA: Designed” and summarized in the “V” column of Tables IV and V.

1) *Inductance*: The d- and q-axis inductances were experimentally measured to calculate the saliency ratio of the prototype. The stator was excited with 60 Hz AC currents injected along each axis, similar to the approach in [34, Sec. VIII]. To do this, a power supply was connected to the stator winding using the circuit diagrams of Fig. 16, with the rotor locked into a position such that Fig. 14(b) creates a d-axis current.

Equation (5) was used to calculate the d- and q-axis inductances using RMS phase voltage $V_{a,\text{rms}}$, RMS phase current

$I_{a,\text{rms}}$, speed ω , and phase angles $\sin(\psi_{v,i})$.

$$L_{d,q} = \frac{V_{a,\text{rms}}}{\omega I_{a,\text{rms}}} \sin(\psi_{v,i}) \quad (5)$$

The results are presented in Table VI. While the experimental measurements are significantly mismatched against FEA of the as-designed rotor (35.6% lower L_d , a 11.8% higher L_q , leading to a 42.1% lower saliency ratio), they are reasonably correlated with the as-printed model (<18% error in L_d , <5% error in L_q). This indicates that the low inductances can be attributed to the printing defects in the prototype, as opposed to shortcomings of the model assumptions. Unfortunately, Table VI shows that the printing defects result in the prototype having an $(L_d - L_q)$ value approximately 49% lower than the design FEA which (1) indicates will result in a similar reduction in torque capability for the prototype.

2) *Torque*: To measure the torque profile, the stator is excited using DC currents along the q-axis (Fig. 16(b)). The rotor is held in different positions in increments of one degree over a rotation. At each position, the reaction torque is measured using the load cell. This test is repeated at three different current values, up to the rated current of the machine.

Fig. 17(a) and (b) compare the measured torque data with FEA calculations. These waveforms illustrate the torque reduction (as compared to the as-designed FEA) due to the printing defects as well as the increased accuracy of the as-printed FEA. In particular, Fig. 17(b) shows that these defects have resulted in a slightly asymmetric torque waveform, also captured by the as-printed FEA, where one of the poles produces a lower torque peak. The torque results are post-processed to analyze their harmonic content, and presented in Fig. 17(c) and (d). The rotor clearly exhibits the typical torque profile of a SynR motor in Fig. 17(c) and the harmonic spectra in Fig. 17(d) further illustrates the ability of the as-printed FEA model to capture the rotor’s torque behavior.

These results show that the prototype machine has a rated torque of 4.15 Nm, 31% lower than the rated torque of the FEA design. Even with the compromised torque rating, the results provide evidence of MMAM technology’s potential to create SynR motors with leading power density values when fabrication shortcomings are resolved to prevent cracking. At the design speed (57,473 RPM, see Table IV), this torque would give the machine a rated power of 25 kW and a power density

of 10.7 kW/L, which is significantly higher than conventional motors (see Table V).

VIII. CONCLUSION

This paper investigated the use of MMAM to create power-dense, rare-earth material-free electric machines. It proposed, printed, and tested a solid SynR rotor. The core innovation is in the use of new MMAM developments to enable the substitution of air in rotor flux barriers with solid, non-magnetic steel, which leads to enhanced rotor strength, facilitating higher tip speed operation and increased power densities.

This paper first focused on refining the MMAM printing process to fabricate high preforming test samples of Fe-3.5Si and 316 L stainless steel. It was found that these materials can be printed to create an interfacial bond stronger than the yield strength of the weaker material (316L) and that the printed Fe-3.5Si has magnetic performance comparable to M19 29 Gauge steel. It is concluded that MMAM technology can be used to print a rotor that has the structural integrity of a solid steel cylinder with the magnetic characteristics of a SynR rotor.

New SynR rotor concepts were then proposed to take advantage of MMAM and a modeling and optimization framework was developed to explore their design space. Two of the new rotor concepts were constrained by limitations of the current state-of-art MMAM technology while an additional “vision” rotor was explored with no limitations to understand the performance potential of this concept once MMAM capabilities have matured. From this study, the paper finds that printing solid MMAM rotors enables SynR motors to achieve a 4x increase in tip speed (> 300 m/s) and power density (> 12 kW/L), while maintaining high efficiencies. Even the rotor concepts constrained by today’s MMAM capabilities were able to achieve power density ranges higher than values reported for other conventional electric machine topologies that use standard materials and cooling.

An optimal rotor design was selected and fabricated. This is the first prototype published of a printed MMAM SynR rotor. While the authors’ ad hoc MMAM equipment was found to introduce structural defects that limited the magnetic performance and maximum rotational speed, the prototype was still a functional SynR machine. It exhibited a saliency ratio of 2.59 and a torque rating of 4.15 Nm. This corresponds to a power density of 10.7 kW/L at the rotor’s rated speed of 57,473 r/min. A pathway to overcome these structural issues to enable a future print that is capable of the design’s rated speed was identified. The paper shows that this pathway can be implemented by leveraging advancements in upcoming commercial MMAM printing components.

Finally, in conclusion, this paper demonstrates the immense potential of MMAM technology to enhance SynR speed and power density. Leveraging commercially available MMAM systems, that bear none of the limitations of the ad hoc system used in this study, could revolutionize the electric machines industry by enabling the production of high-speed, power-dense SynR rotors without relying on rare-earth permanent magnets, paving the way for a more sustainable future.

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